

Applications

in Chemoinformatics

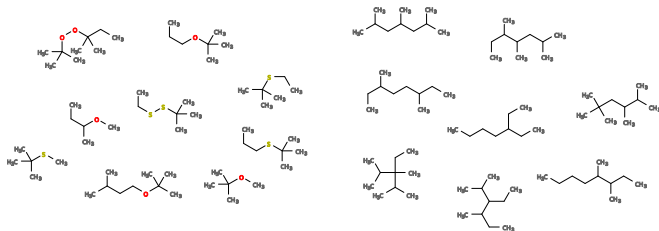
- ▶ Boiling point prediction
- ▶ Electrophilicity
- ▶ Redox Potential

Predictive performance of graph kernels

Boiling point prediction

- ▶ Acyclic : 183 graphs, with heteroatoms, no cycles
- ▶ Alkane : 150 graphs, only carbons
- ▶ RMSE in °C

Method	Acyclic	Alkane
Linear Patterns	9.00 ± 12.87	6.66 ± 1.63
Treelet Kernel	6.99 ± 1.45	2.53 ± 1.32
WL subtree	19.80 ± 6.12	26.42 ± 41.59



Predicting electrophilicity from tabular data I

Joint work with chemists

- ▶ Dataset of 111 molecules with electrophilicity
- ▶ Quantum descriptors
- ▶ Curated by COBRA (Rouen)

SOTA

- ▶ [González et al., 2018]
- ▶ Genetic Programming
- ▶ No protocol

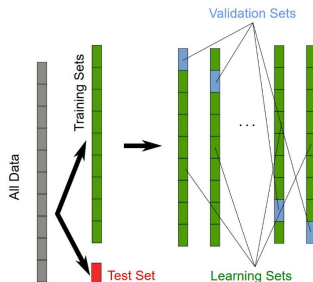
$$E_{\text{Mayr}} = -22.45682 + 0.47192 \left[-\frac{4.18726}{A} - \frac{17.83630}{-3.90327 + 1.54873A} - 0.27450A^2 \times (-5.61841 - 0.37166A + 0.311821) + 1.394661 \right] \quad (28)$$

Predicting electrophilicity from tabular data II

Our contribution

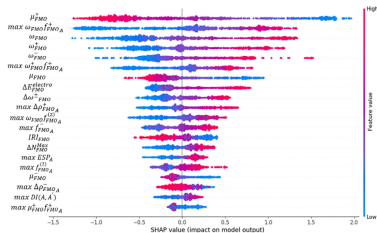
[Hoffmann et al., 2020]

- ▶ Transform graph into tabular data
- ▶ Three levels of descriptors :
 - ▶ Graph level
 - ▶ Topological level
 - ▶ Atom level
- ▶ Classic ML methods
- ▶ Clear protocol



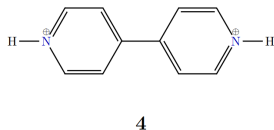
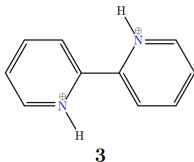
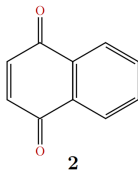
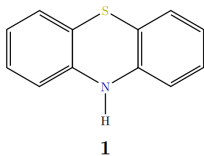
The results

- ▶ Improved accuracy
- ▶ Explainable model



Redox

- ▶ Prediction of reduction/oxydation potential (ΔG)
 - ▶ Reduction : ΔG_{red}^{PBE0}
 - ▶ Oxydation : ΔG_{red}^{PBE0}
- ▶ Dataset : 318 reactions for 159 compounds
- ▶ Experimental data : Expensive data
- ▶ Collaboration with COBRA and Éric Brémond (ITODYS)



Results on Redox

Models	ΔG_{red}^{PBE0}			ΔG_{ox}^{PBE0}		
	Train	Valid	Test	Train	Valid	Test
Random	0.0±0.0	110.5±14.3	108.0±17.2	0.0±0.0	118.1±24.9	122.8±16.7
Mean	0.0±0.0	97.6±4.8	99.2±3.5	0.0±0.0	95.2±6.0	96.7±5.3
LR (af/stats)	-	-	∞	-	-	∞
LR (af/count)	22.9±1.0	28.2±6.2	28.7±4.1	23.9±0.99	29.6±6.2	30.3±3.9
GPR (af/stats)	10.4±5.5	19.4±3.9	23.3±6.1	11.0±5.4	20.2±5.6	24.3±4.8
GPR (af/count)	19.4±3.6	24.4±5.4	25.8±4.9	21.6±3.7	26.7±6.0	30.2±5.8
GKRR (af/stats)	5.5±7.7	17.9±4.5	23.0±6.6	7.8±5.0	18.3±5.0	24.0±4.7
GKRR (af/count)	5.0±4.5	12.6±4.0	17.1±5.0	5.5±5.1	15.5±4.8	20.4±6.7
SVR (af/stats)	6.3±4.7	16.6±4.6	22.2±6.3	8.6±5.3	17.3±5.4	23.3±5.1
SVR (af/count)	3.1±2.7	12.2±3.8	15.1±5.1	2.7±3.1	14.0±4.1	18.2±6.3
RF (af/stats)	7.1±1.0	14.4±4.5	15.7±4.9	7.5±1.7	15.6±5.1	18.3±6.3
RF (af/count)	5.9±1.9	11.8±4.7	13.9±5.3	6.2±1.1	13.9±4.2	16.4±5.8
XGB (af/stats)	0.5±0.3	9.8±4.1	12.3±6.0	0.6±0.5	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±1.0	9.5±4.5	13.8±4.2	1.1±1.0	11.4±5.2	16.2±5.9
KNN (af/stats)	2.1±4.4	13.8±6.7	18.5±9.0	1.5±4.2	16.2±5.9	20.2±7.6
KNN (af/count)	1.5±3.0	7.9±2.8	10.3±3.7	0.9±2.2	10.1±3.7	13.0±6.3
GED (BP-random-types)	1.1±2.9	12.2±10.1	14.7±7.0	1.4±3.6	14.0±6.0	15.8±6.8
GED (BP-random-feats/ δ)	1.3±3.0	9.1±5.2	12.1±5.8	1.7±4.0	12.0±5.3	14.7±5.3
GED (BP-random-feats/dis)	1.8±3.6	7.8±4.9	10.8±7.2	2.4±4.3	11.3±4.7	14.4±5.0
GED (BP-random-afeats/dis+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	0.5±1.9	12.4±5.0	16.1±7.4
GED (BP-fitted-types)	0.4±2.1	8.0±4.0	10.8±4.1	1.4±4.5	12.2±4.7	15.5±5.2
GED (BP-fitted-feats/ δ)	0.4±1.6	7.1±2.5	8.9±3.0	1.8±4.2	10.3±3.8	13.3±4.3
GED (BP-fitted-feats/dis)	0.5±1.5	6.7±2.4	7.8±2.7	0.6±2.4	10.2±3.4	12.1±4.1
GED (BP-fitted-afeats/dis+3d/dis)	0.5±1.4	8.0±3.8	9.8±4.0	1.4±3.3	12.0±5.9	13.4±5.4
MPNN + transformer	3.6±1.7	4.7±2.0	10.0±4.3	3.9±1.8	6.7±3.0	14.3±5.7
GCN	4.2±1.9	5.3±2.2	10.1±3.5	4.6±2.9	6.7±2.4	12.9±4.4
GAT	4.5±2.1	5.3±2.2	10.0±2.7	4.8±2.1	7.4±3.0	14.2±3.6
DSGCN	4.8±3.5	5.9±2.7	10.7±3.7	7.0±6.5	7.4±3.4	14.4±5.5
EGNN (+ node descs)	4.0±2.2	3.4±1.4	9.7±5.3	4.4±1.5	4.2±2.0	11.5±5.6

Results on Redox

Models	ΔG_{red}^{PBE0}			ΔG_{ox}^{PBE0}		
	Train	Valid	Test	Train	Valid	Test
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RF (af/count)	5.9±1.9	11.8±4.7	13.9±5.3	6.2±1.1	13.9±4.2	16.4±5.8
XGB (af/stats)	0.5±0.3	9.8±4.1	12.3±6.0	0.6±0.5	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±1.0	9.5±4.5	13.8±4.2	1.1±1.0	11.4±5.2	16.2±5.9
KNN (af/stats)	2.1±4.4	13.8±6.7	18.5±9.0	1.5±4.2	16.2±5.9	20.2±7.6
KNN (af/count)	1.5±3.0	7.9±2.8	10.3±3.7	0.9±2.2	10.1±3.7	13.0±6.3
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GED (BP-random-feats/dis)	1.8±3.6	7.8±4.9	10.8±7.2	2.4±4.3	11.3±4.7	14.4±5.0
GED (BP-random-afeats/dis+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	0.5±1.9	12.4±5.0	16.1±7.4
GED (BP-fitted-types)	0.4±2.1	8.0±4.0	10.8±4.1	1.4±4.5	12.2±4.7	15.5±5.2
GED (BP-fitted-feats/ δ)	0.4±1.6	7.1±2.5	8.9±3.0	1.8±4.2	10.3±3.8	13.3±4.3
GED (BP-fitted-feats/dis)	0.5±1.5	6.7±2.4	7.8±2.7	0.6±2.4	10.2±3.4	12.1±4.1
GED (BP-fitted-afeats/dis+3d/dis)	0.5±1.4	8.0±3.8	9.8±4.0	1.4±3.3	12.0±5.9	13.4±5.4
MPNN + transformer	3.6±1.7	4.7±2.0	10.0±4.3	3.9±1.8	6.7±3.0	14.3±5.7
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GAT	4.5±2.1	5.3±2.2	10.0±2.7	4.8±2.1	7.4±3.0	14.2±3.6
DSGCN	4.8±3.5	5.9±2.7	10.7±3.7	7.0±6.5	7.4±3.4	14.4±5.5
EGNN (+ node descs)	4.0±2.2	3.4±1.4	9.7±5.3	4.4±1.5	4.2±2.0	11.5±5.6

Performances

Results on Redox

Models	ΔG_{red}^{PBE0}			ΔG_{ox}^{PBE0}		
	Train	Valid	Test	Train	Valid	Test
Random	0.0±0.0	110.5±14.3	108.0±17.2	0.0±0.0	118.1±24.9	122.8±16.7
Mean	0.0±0.0	97.6±4.8	99.2±3.5	0.0±0.0	95.2±6.0	96.7±5.3
LR (af/stats)	-	-	∞	-	-	∞
LR (af/count)	22.9±1.0	28.2±6.2	28.7±4.1	23.9±0.99	29.6±6.2	30.3±3.9
GPR (af/stats)	10.4±5.5	19.4±3.9	23.3±6.1	11.0±5.4	20.2±5.6	24.3±4.8
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GKRR (af/stats)	5.5±7.7	17.9±4.5	23.0±6.6	7.8±5.0	18.3±5.0	24.0±4.7
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SVR (af/stats)	6.3±4.7	16.6±4.6	22.2±6.3	8.6±5.3	17.3±5.4	23.3±5.1
SVR (af/count)	3.1±2.7	12.2±3.8	15.1±5.1	2.7±3.1	14.0±4.1	18.2±6.3
RF (af/stats)	7.1±1.0	14.4±4.5	15.7±4.9	7.5±1.7	15.6±5.1	18.3±6.3
RF (af/count)	5.9±1.9	11.8±4.7	13.9±5.3	6.2±1.1	13.9±4.2	16.4±5.8
XGB (af/stats)	0.5±0.3	9.8±4.1	12.3±6.0	0.6±0.5	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±1.0	9.5±4.5	13.8±4.2	1.1±1.0	11.4±5.2	16.2±5.9
KNN (af/stats)	2.1±4.4	13.8±6.7	18.5±9.0	1.5±4.2	16.2±5.9	20.2±7.6
KNN (af/count)	1.5±3.0	7.9±2.8	10.3±3.7	0.9±2.2	10.1±3.7	13.0±6.3
GED (BP-random-types)	1.1±2.9	12.2±10.1	14.7±7.0	1.4±3.6	14.0±6.0	15.8±6.8
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GED (BP-random-feats/dis)	1.8±3.6	7.8±4.9	10.8±7.2	2.4±4.3	11.3±4.7	14.4±5.0
GED (BP-random-afeats/dis+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	0.5±1.9	12.4±5.0	16.1±7.4
GED (BP-fitted-types)	0.4±2.1	8.0±4.0	10.8±4.1	1.4±4.5	12.2±4.7	15.5±5.2
GED (BP-fitted-feats/ δ)	0.4±1.6	7.1±2.5	8.9±3.0	1.8±4.2	10.3±3.8	13.3±4.3
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MPNN + transformer	3.6±1.7	4.7±2.0	10.0±4.3	3.9±1.8	6.7±3.0	14.3±5.7
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GAT	4.5±2.1	5.3±2.2	10.0±2.7	4.8±2.1	7.4±3.0	14.2±3.6
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EGNN (+ node descs)	4.0±2.2	3.4±1.4	9.7±5.3	4.4±1.5	4.2±2.0	11.5±5.6

Results on Redox

Models	ΔG_{red}^{PBE0}			ΔG_{ox}^{PBE0}		
	Train	Valid	Test	Train	Valid	Test
Random	0.0±0.0	110.5±14.3	108.0±17.2	0.0±0.0	118.1±24.9	122.8±16.7
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XGB (af/stats)	0.5±0.3	9.8±4.1	12.3±6.0	0.6±0.5	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±1.0	9.5±4.5	13.8±4.2	1.1±1.0	11.4±5.2	16.2±5.9
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Simple KNN
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LR (af/count)	22.9±1.0	28.2±6.2	28.7±4.1	23.9±0.99	29.6±6.2	30.3±3.9
GPR (af/stats)	10.4±5.5	19.4±3.9	23.3±6.1	11.0±5.4	20.2±5.6	24.3±4.8
GPR (af/count)	19.4±3.6	24.4±5.4	25.8±4.9	21.6±3.7	26.7±6.0	30.2±5.8
GKRR (af/stats)	5.5±7.7	17.9±4.5	23.0±6.6	7.8±5.9	18.3±5.0	24.0±4.7
GKRR (af/count)	5.0±4.5	16.1±4.0	21.5±5.1	5.5±4.1	16.5±4.8	20.4±6.7
SVR (af/stats)	6.3±4.7	16.1±4.0	20.9±5.1	6.9±4.9	16.3±5.4	23.3±5.1
SVR (af/count)	3.1±2.7	12.2±3.8	15.1±5.1	2.7±3.1	14.0±4.1	18.2±6.3
RF (af/stats)	7.1±1.0	14.4±4.5	15.7±4.9	7.5±1.7	15.6±5.1	18.3±6.3
RF (af/count)	5.9±1.9	11.8±4.7	14.0±5.0	5.7±1.9	13.9±4.2	16.4±5.8
XGB (af/stats)	0.5±0.3	9.8±4.1	12.0±4.0	0.5±0.3	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±1.0	9.5±4.5	13.8±4.2	1.1±1.0	11.4±5.2	16.2±5.9
KNN (af/stats)	2.1±4.4	13.8±4.1	15.9±4.9	1.5±4.2	16.2±5.0	20.2±7.6
KNN (af/count)	0.9±0.9	11.1±4.0	14.1±4.3	0.9±0.9	13.6±4.3	16.4±5.3
GED (BP-random-types)	1.2±2.9	14.8±10.1	14.7±7.0	1.4±3.6	14.0±6.0	15.8±6.8
GED (BP-random-feats/ δ)	1.3±3.0	9.1±5.2	12.1±5.8	1.7±4.0	12.0±5.3	14.7±5.3
GED (BP-random-feats/dis)	1.8±3.6	7.8±4.9	10.8±7.2	2.4±4.3	11.3±4.7	14.4±5.0
GED (BP-random-afeats/dis+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	0.5±1.9	12.4±5.0	16.1±7.4
GED (BP-fitted-types)	0.4±2.1	8.0±4.0	10.8±4.1	1.4±4.5	12.2±4.7	15.5±5.2
GED (BP-fitted-feats/ δ)	0.4±1.6	7.1±2.5	8.9±3.0	1.8±4.2	10.3±3.8	13.3±4.3
GED (BP-fitted-feats/dis)	0.5±1.5	6.7±2.4	7.8±2.7	0.6±2.4	10.2±3.4	12.1±4.1
GED (BP-fitted-afeats/dis+3d/dis)	0.5±1.4	8.0±3.8	9.8±4.0	1.4±3.3	12.0±5.9	13.4±5.4
MPNN + transformer	3.6±1.7	4.7±2.0	10.0±4.3	3.9±1.8	6.7±3.0	14.3±5.7
GCN	4.2±1.9	5.3±2.2	10.1±3.5	4.6±2.9	6.7±2.4	12.9±4.4
GAT	4.5±2.1	5.3±2.2	10.0±2.7	4.8±2.1	7.4±3.0	14.2±3.6
DSGCN	4.8±3.5	5.9±2.7	10.7±3.7	7.0±6.5	7.4±3.4	14.4±5.5
EGNN (+ node descs)	4.0±2.2	3.4±1.4	9.7±5.3	4.4±1.5	4.2±2.0	11.5±5.6

Results on Redox

Models	ΔG_{red}^{PBE0}			ΔG_{ox}^{PBE0}		
	Train	Valid	Test	Train	Valid	Test
Random	0.0±0.0	110.5±14.3	108.0±17.2	0.0±0.0	118.1±24.9	122.8±16.7
Mean	0.0±0.0	97.6±4.8	99.2±3.5	0.0±0.0	95.2±6.0	96.7±5.3
LR (af/stats)	-	-	∞	-	-	∞
LR (af/count)	22.9±1.0	28.2±6.2	28.7±4.1	23.9±0.99	29.6±6.2	30.3±3.9
GPR (af/stats)	10.4±5.5	19.4±3.9	23.3±6.1	11.0±5.4	20.2±5.6	24.3±4.8
GPR (af/count)	19.4±3.6	24.4±5.4	25.8±4.9	21.6±3.7	26.7±6.0	30.2±5.8
GKRR (af/stats)	5.5±7.7	17.0±4.5	23.0±6.6	7.8±5.0	18.3±5.0	24.0±4.7
GKRR (af/count)	10.1±5.5	2.6±4.0	17.1±5.0	5±5.1	15.5±4.8	20.4±6.7
SVR (af/stats)	10.4±4.4	10.1±4.6	22.1±4.7	8.1±5.3	12.3±4.4	23.3±5.1
SVR (af/count)	3.4±2.2	1.8±4.8	15.7±4.9	4.3±5.3	11.6±4.4	18.2±6.3
RF (af/stats)	7.1±1.0	1.1±4.5	15.7±4.9	6.2±1.7	11.6±4.4	18.3±6.3
RF (af/count)	5.9±1.9	11.8±4.7	13.9±5.3	6.2±1.1	13.9±4.2	16.4±5.8
XGB (af/stats)	0.5±0.3	9.8±4.1	12.3±6.0	0.6±0.5	12.1±5.2	16.5±5.6
XGB (af/count)	1.1±0.3	13.8±4.9	13.8±6.0	1.1±0.6	11.4±5.2	16.2±5.9
KNN (af/stats)	2.0±0.3	1.5±6.1	10.3±3.7	1.1±0.6	16.2±5.9	20.2±7.6
KNN (af/count)	1.2±0.0	7.9±5.8	10.3±3.7	0.9±2.2	10.1±3.7	13.0±6.3
GED (BP-random-types)	1.1±2.9	12.2±10.1	14.7±7.0	1.4±3.6	14.0±6.0	15.8±6.8
GED (BP-random-feats+6d/dis)	1.3±2.9	9.9±5.5	12.1±5.6	1.7±3.6	12.9±5.9	15.5±5.3
GED (BP-random-feats+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	2.4±3.6	10.4±4.1	13.4±4.1
GED (BP-random-af-feats/dis+3d/dis)	0.0±0.0	10.4±9.0	10.9±8.0	0.5±1.9	12.4±3.0	10.1±7.7
GED (BP-fitted-types)	0.4±2.1	8.0±4.0	10.8±4.1	1.4±4.5	12.2±4.7	15.5±5.2
GED (BP-fitted-feats/δ)	0.4±1.6	7.1±2.5	8.9±3.0	1.8±4.2	10.3±3.8	13.3±4.3
GED (BP-fitted-feats/dis)	0.5±1.5	6.7±2.4	7.8±2.7	0.6±2.4	10.2±3.4	12.1±4.1
GED (BP-fitted-af-feats/dis+3d/dis)	0.5±1.4	8.0±3.8	9.8±4.0	1.4±3.3	12.0±5.9	13.4±5.4
MPNN + transformer	3.6±1.7	4.7±2.0	10.0±4.3	3.9±1.8	6.7±3.0	14.3±5.7
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DSGCN	4.8±3.5	5.9±2.7	10.7±3.7	7.0±6.5	7.4±3.4	14.4±5.5
EGNN (+ node descs)	4.0±2.2	3.4±1.4	9.7±5.3	4.4±1.5	4.2±2.0	11.5±5.6

GNNs combine
graph and
representation learning